

JUNO Reactor Antineutrino + Neutron Monte Carlo

Progressive simulation report — Improved Physics

Detector mass: 20.0 kt

Generated MC events: 500,000

Selected IBD-like events: 492,665

Selection efficiency: 0.9853

Estimated free protons: $1.445\text{e}+33$

Oscillation hierarchy used for MC: NO

Note: Huber-Mueller reactor spectrum, improved IBD cross section, and material-dependent neutron scattering.

Simulation Configuration

Simulation configuration

Scintillator mass = 20.00 kt

Target radius = 17.70 m

Energy resolution coefficient = 0.0302

Light yield = 1345.0 PE/MeV

Hydrogen-target approximation = 12% mass fraction

Neutron capture time constant = 200.0 us

Neutron capture gamma energy = 2.223 MeV

Reactor configuration

Yangjiang: 17.40 GWth, 52.80 km

Taishan: 9.20 GWth, 53.10 km

Fission fractions

U235: 0.550

U238: 0.080

Pu239: 0.300

Pu241: 0.070

IBD selection

Prompt: 0.70-8.00 MeV

Delayed: 1.80-2.60 MeV

Time window: 1000.0 us

Spatial cut: 2.00 m

Physics improvements:

- Huber-Mueller reactor spectrum (polynomial parameterization)
- IBD cross section with recoil and radiative corrections
- Neutron thermalization with H and C elastic scattering cross sections
- Capture on hydrogen using 1/v cross section

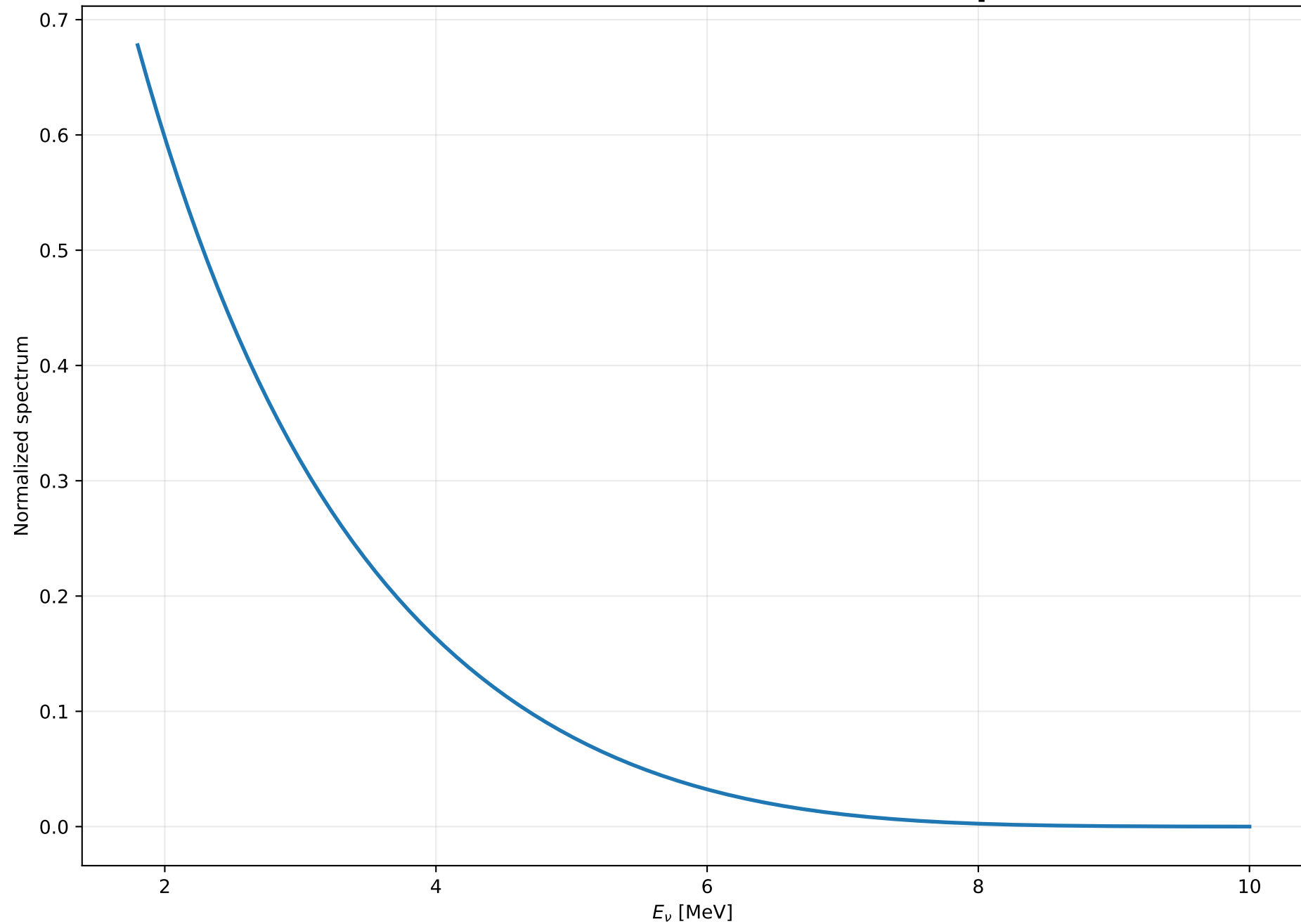
Monte Carlo Result Summary

Quantity	Value
Generated events	500000
Selected events	492665
Selection efficiency	0.98533
Mean neutrino energy [MeV]	5.17067
Mean prompt energy [MeV]	4.38835
Mean delayed energy [MeV]	2.22297
Mean neutron initial energy [keV]	60.9491
Mean capture time [us]	199.996
Mean capture displacement [cm]	0.282788
Estimated free protons	1.44531e+33

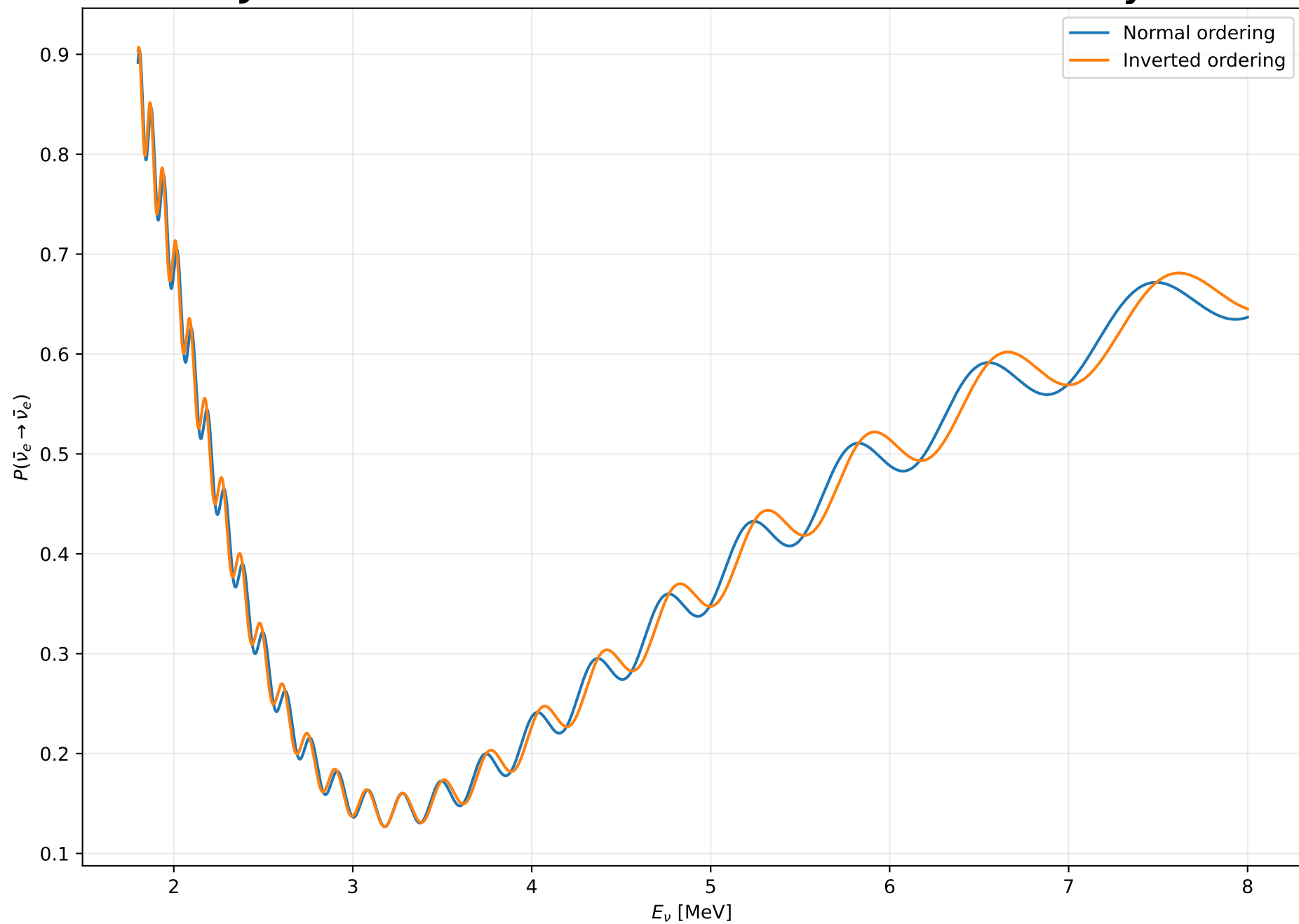
Sample of Generated Events

E_nu_MeV	E_prompt_reco_MeV	E_delayed_reco_MeV	neutron_initial_MeV	capture_time_us	capture_distance_m
5.746201	4.869888	2.171747	0.087405	3.171592	0.000245
5.094925	4.324164	2.260223	0.10772	49.391071	0.000467
6.407662	5.57026	2.272862	0.084177	96.708217	0.000945
4.839192	4.101115	2.19777	0.032083	195.84302	0.00223
5.670179	4.843123	2.179182	0.035674	63.400166	0.002912
4.507364	3.729368	2.25948	0.040494	313.55103	0.003754
2.914796	2.107807	2.234944	0.028017	10.014043	0.000607
5.966768	5.133086	2.194796	0.078631	50.89884	0.001019
2.067121	1.324164	2.191078	0.01044	96.157848	0.002627
5.655381	4.979926	2.239405	0.034379	39.821814	0.001098
6.378229	5.607435	2.233457	0.031934	15.900607	0.00113
4.037109	3.233457	2.209665	0.088344	535.83619	0.006196

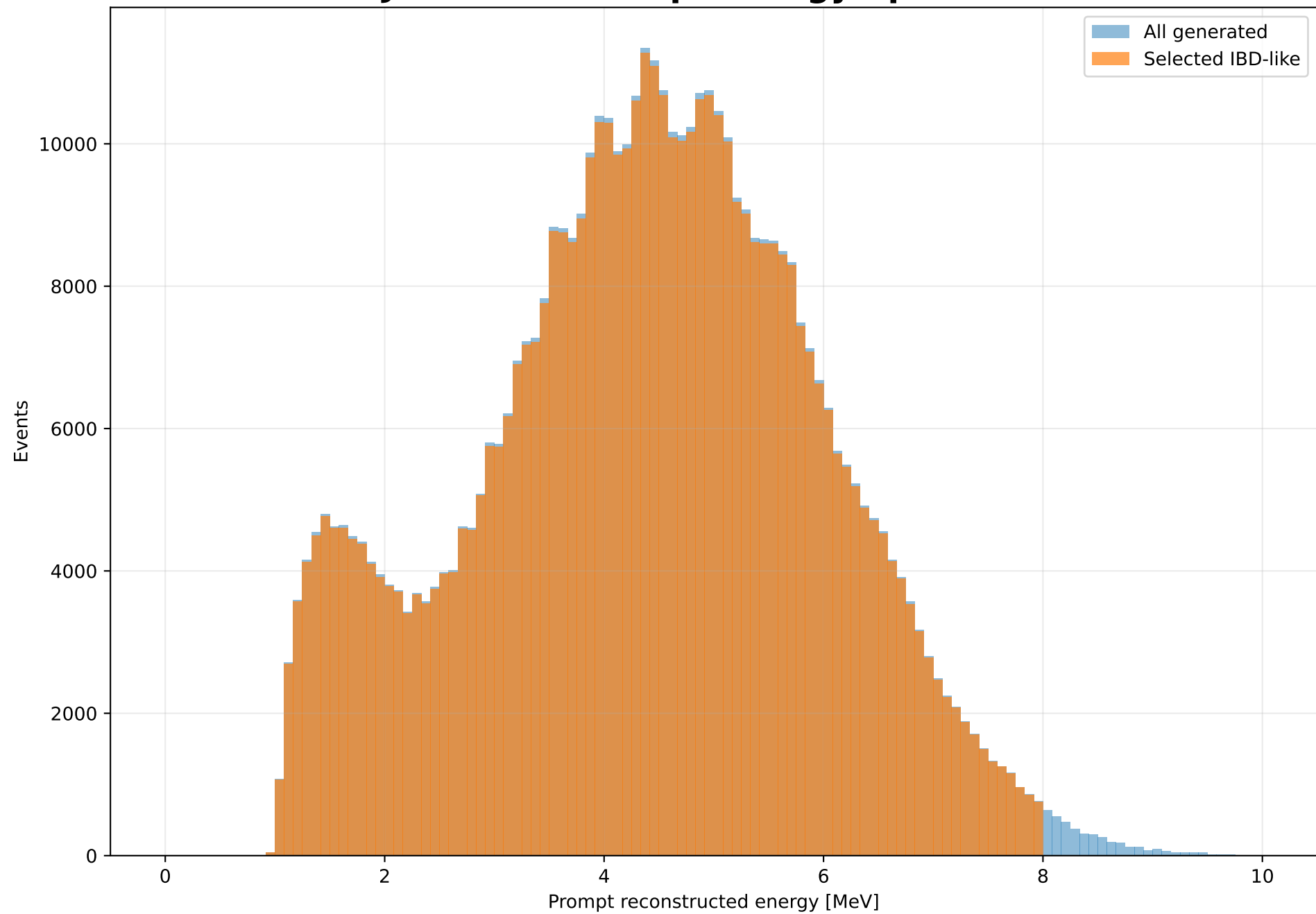
Huber-Mueller Reactor Antineutrino Spectrum



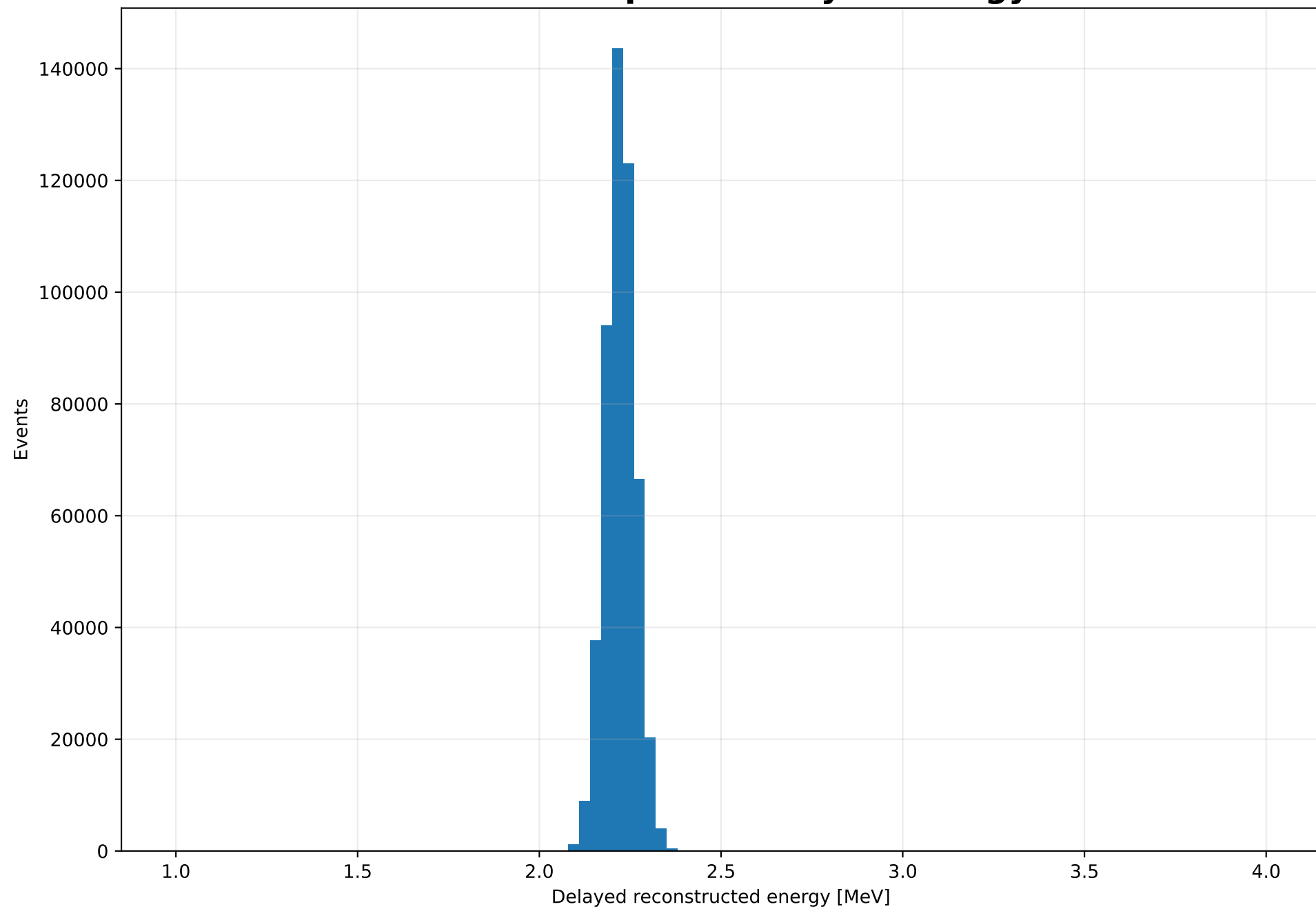
JUNO Electron Antineutrino Survival Probability



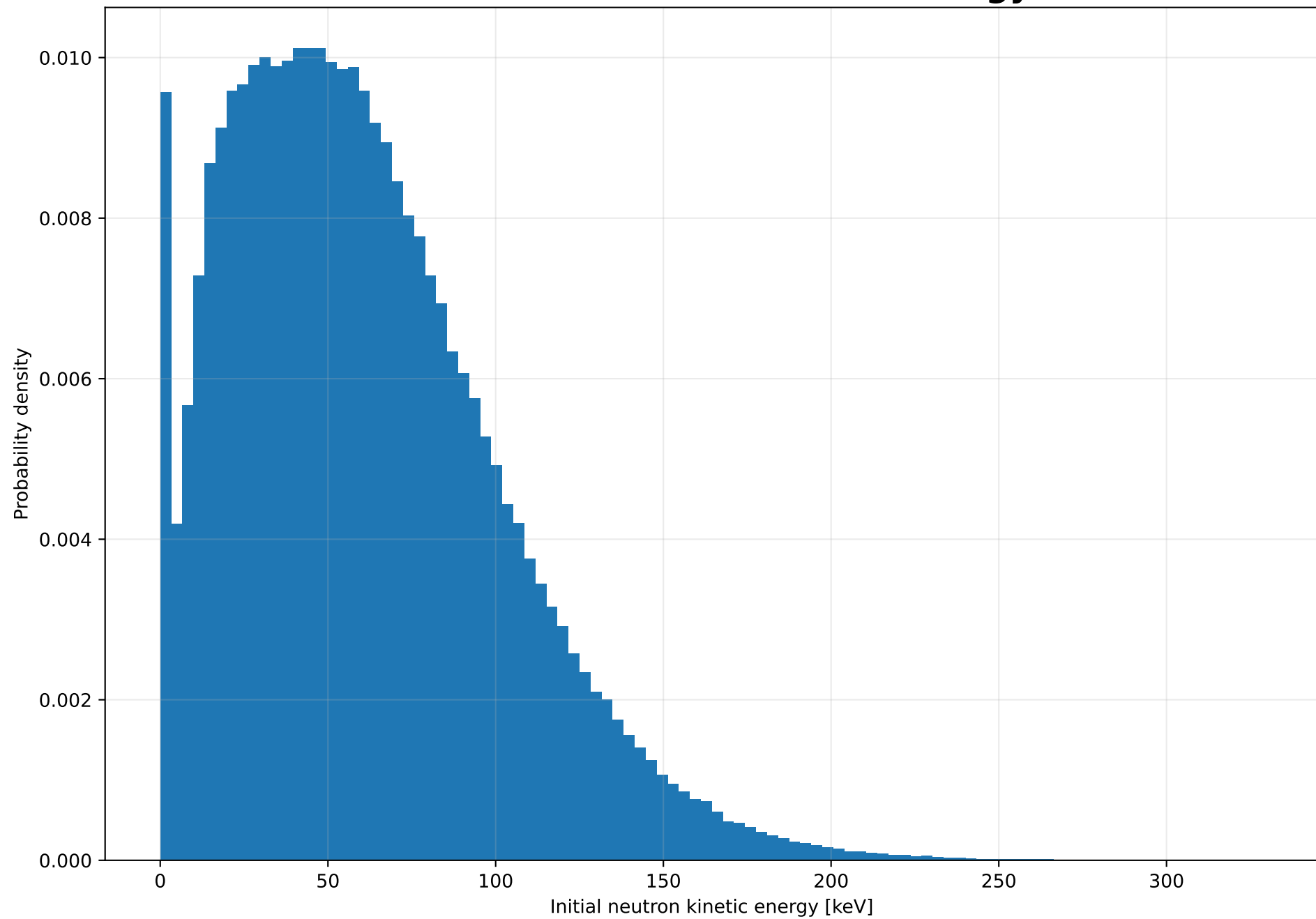
JUNO-like Prompt Energy Spectrum



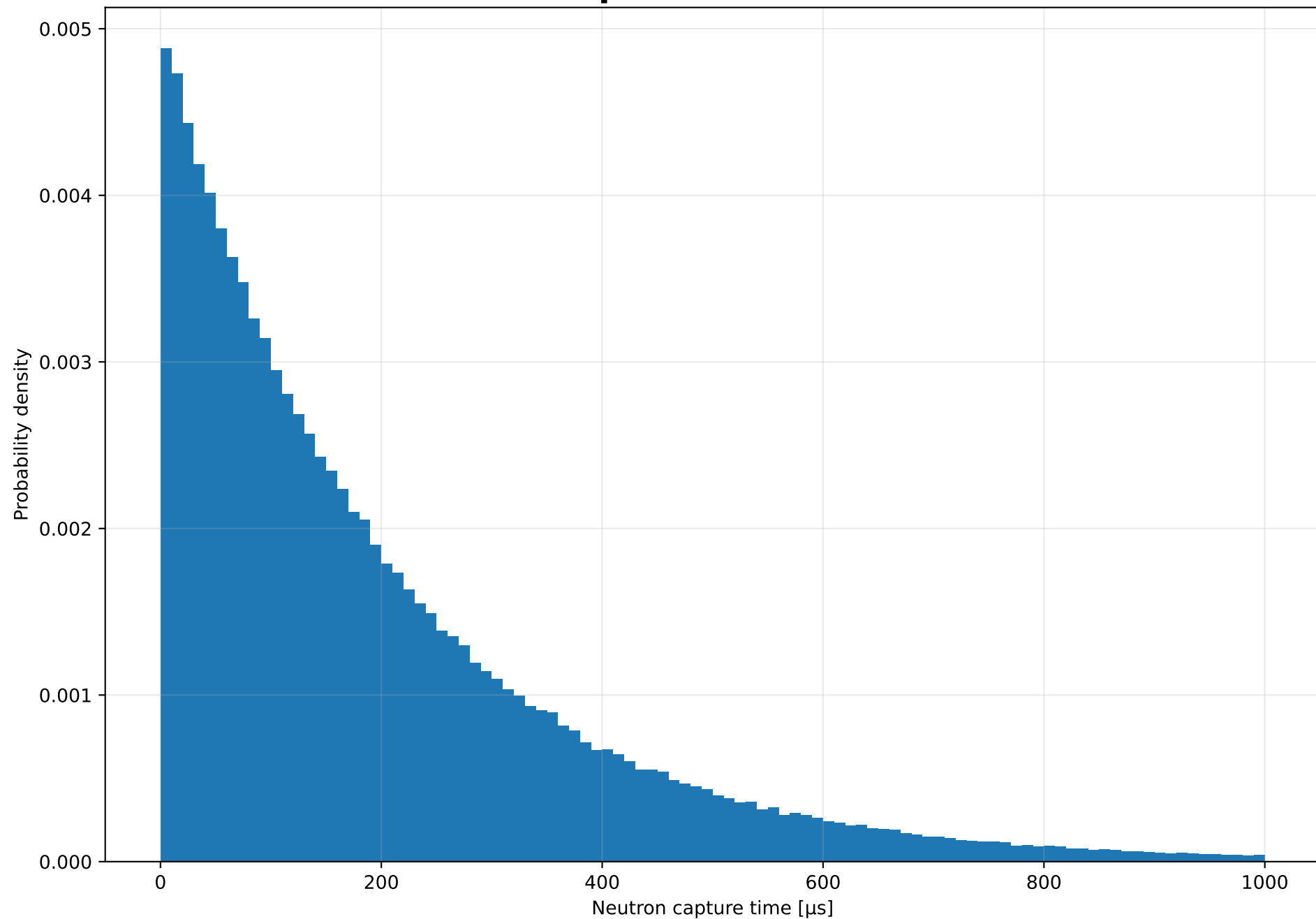
Neutron-Capture Delayed Energy



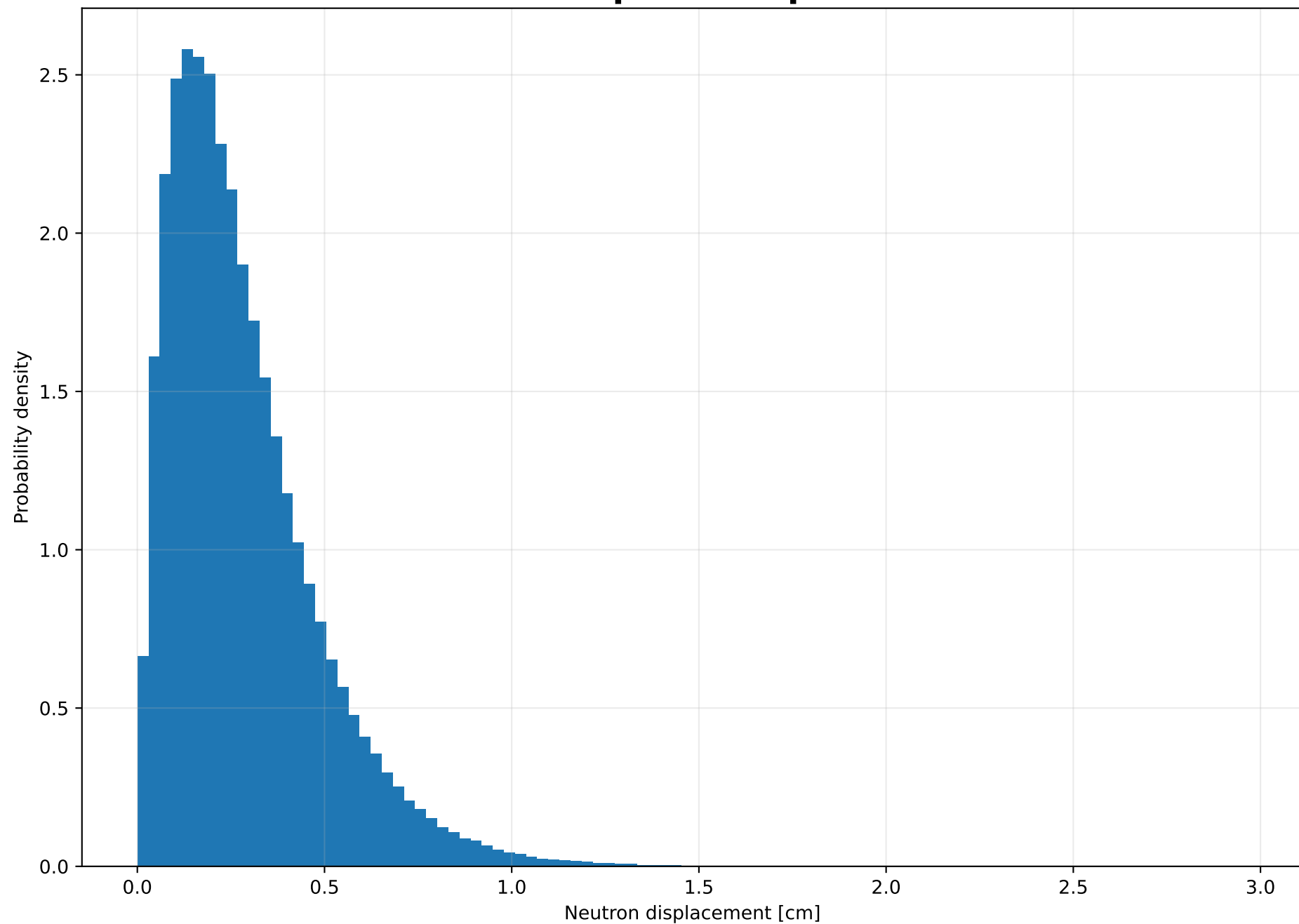
IBD Neutron Initial Kinetic Energy



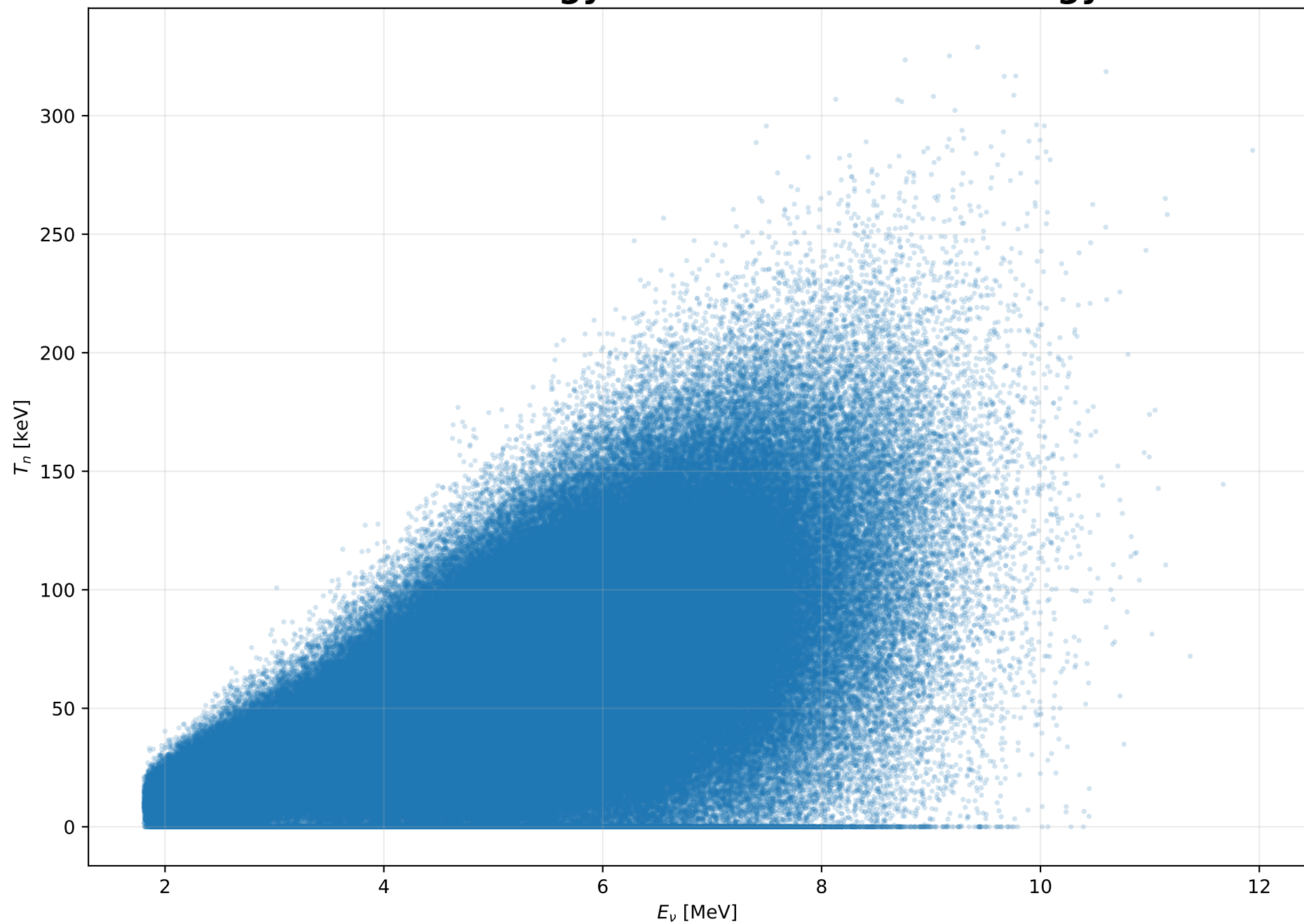
Neutron Capture-Time Distribution



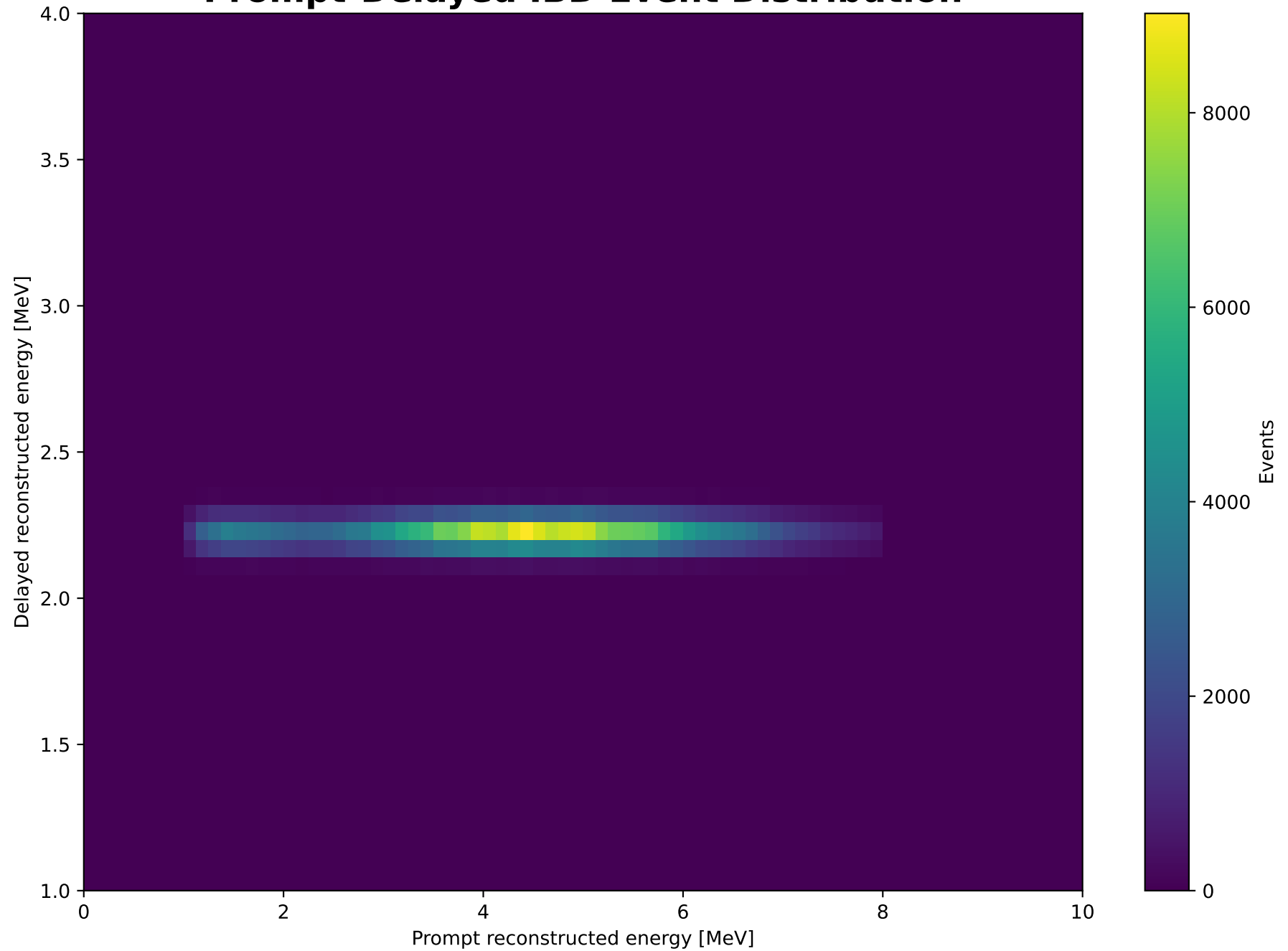
Neutron Capture Displacement



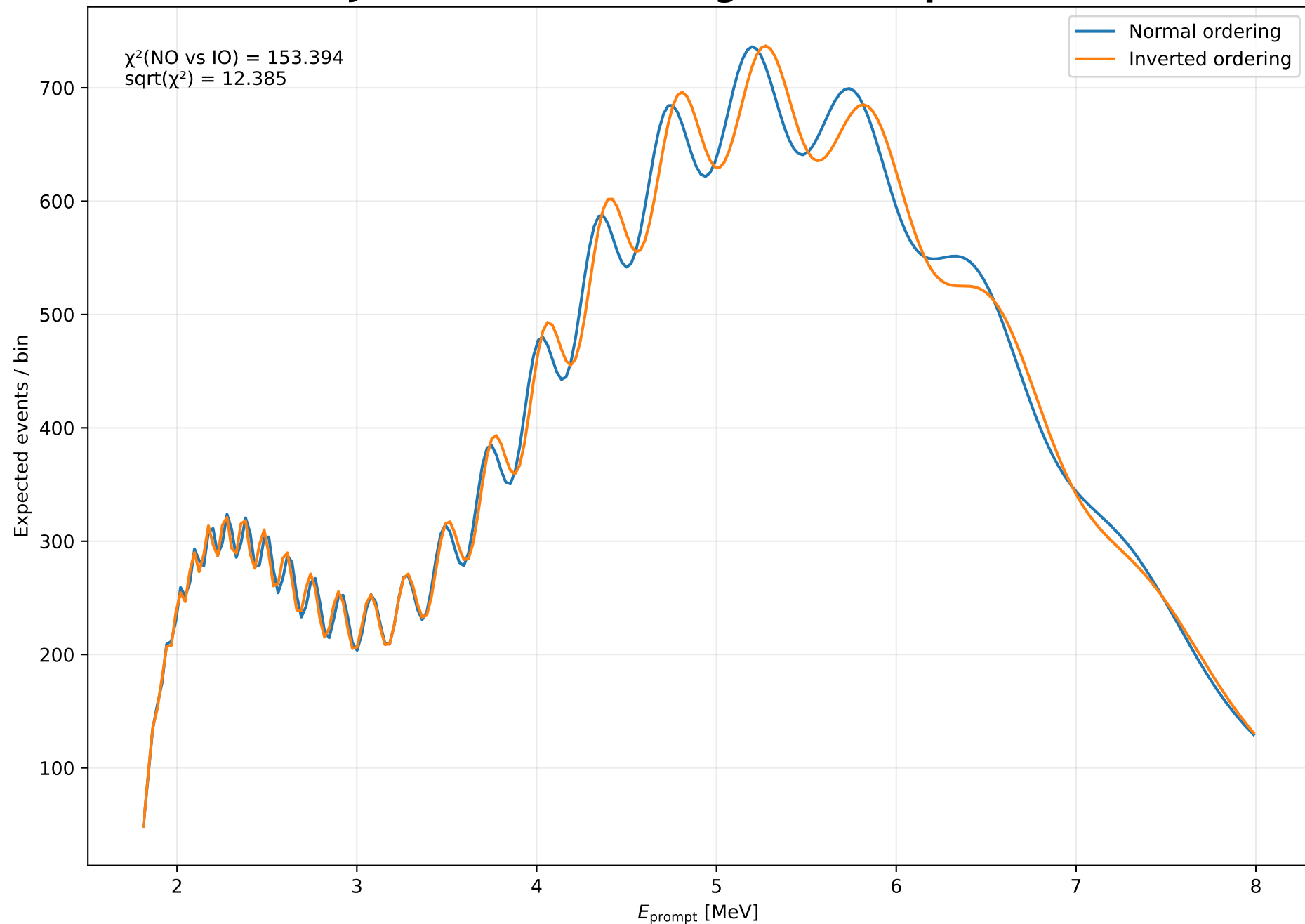
Neutrino Energy vs. Initial Neutron Energy



Prompt-Delayed IBD Event Distribution



JUNO Mass-Ordering Asimov Spectra



Interpretation and Limitations

The simulation is designed as a progressive JUNO/neutrino-physics study.

The reactor spectrum uses the Huber-Mueller polynomial parameterization, which is a good approximation but not the most up-to-date prediction.

The IBD cross section includes recoil and a constant radiative correction, but does not include all higher-order corrections.

The neutron thermalization now uses simplified energy-dependent cross sections for hydrogen and carbon, but still does not perform full spatial transport.

Capture time is sampled from an exponential independent of the slowing-down process.

The detector response uses a simplified JUNO-like photoelectron yield and energy-resolution model.

Mass-ordering discrimination here is an idealized Asimov toy study.

Real JUNO sensitivity requires systematics, nonlinearity, reactor model uncertainties, backgrounds, detector response, and proper fitting.